The Normal Distribution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.03

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Properties of the normal distribution**

State three key properties of a normal distribution's shape.

A normal distribution is: (1) symmetric and bell-shaped, centered at its mean; (2) unimodal (a single peak at the mean); (3) its total area under the curve equals 1, and it never touches the horizontal axis (it extends infinitely in both directions, getting closer and closer to zero).

2 **[Foundational]** • **Target: Properties of the normal distribution**

Explain what the two parameters μ and σ control in a normal distribution's shape.

μ (the mean) controls the location of the center of the distribution (where the peak is). σ (the standard deviation) controls the spread: a larger σ produces a wider, flatter curve, while a smaller σ produces a narrower, taller curve — but both remain symmetric and bell-shaped.

3 **[Foundational]** • **Target: The 68-95-99.7 rule**

Adult male heights are approximately normal with mean 70 inches and standard deviation 3 inches. Using the 68-95-99.7 rule, find the range of heights containing the middle 95% of adult men.

The 95% range spans $\mu \pm 2\sigma = 70 \pm 2(3) = 70 \pm 6$, so from 64 to 76 inches.

4 **[Foundational]** • **Target: The 68-95-99.7 rule**

Using the same distribution, find the approximate percentage of adult men taller than 76 inches.

76 inches is $\mu + 2\sigma$. By the 68-95-99.7 rule, 95% of the data lies within 2 standard deviations, leaving 5% split evenly in the two tails: 2.5% above 76 inches and 2.5% below 64 inches.

5 **[Foundational]** • **Target: Standardizing and using the normal distribution**

Using the same height distribution (mean 70, standard deviation 3), find the z-score for a height of 77.5 inches, and use a standard normal table value of $P(Z < 2.5) \approx 0.9938$ to find the proportion of men shorter than 77.5 inches.

$z = \frac{77.5 - 70}{3} = \frac{7.5}{3} = 2.5$. $P(X < 77.5) = P(Z < 2.5) \approx 0.9938$, or about 99.38% of men are shorter than 77.5 inches.

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6 [Foundational] • Target: Standardizing and using the normal distribution

Find the proportion of men between 65 and 75 inches tall, given $P(Z < 1.67) \approx 0.9525$ and $P(Z < -1.67) \approx 0.0475$.

$z_{65} = \frac{65 - 70}{3} \approx -1.67$. $z_{75} = \frac{75 - 70}{3} \approx 1.67$. $P(65 < X < 75) = P(Z < 1.67) - P(Z < -1.67) \approx 0.9525 - 0.0475 = 0.905$, or about 90.5%.

7 [Foundational] • Target: Visualizing the standard normal distribution

The graph shows the standard normal distribution ($\mu = 0$, $\sigma = 1$), with the area to the left of $z = 1$ shaded (though shading itself is not shown, imagine it visually).

a Using $P(Z < 1) \approx 0.8413$, find the area under the curve to the right of $z = 1$.

$P(Z > 1) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$.

b Explain, using the curve's symmetry, why $P(Z < -1) = P(Z > 1)$. The standard normal curve is symmetric about $z = 0$. By symmetry, the area to the left of -1 is a mirror image of the area to the right of 1, so $P(Z < -1)$ and $P(Z > 1)$ must be equal.

(a) $P(Z > 1) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$.; (b) The standard normal curve is symmetric about $z = 0$. By symmetry, the area to the left of -1 is a mirror image of the area to the right of 1, so $P(Z < -1)$ and $P(Z > 1)$ must be equal.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Working backward: finding values from percentiles

SAT scores are approximately normal with mean 1050 and standard deviation 150. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.

$x = \mu + z\sigma = 1050 + 1.28(150) = 1050 + 192 = 1242$.

9 [Intermediate] • Target: Working backward: finding values from percentiles

Using the same distribution, find the score that separates the bottom 10% of scores, given that $z \approx -1.28$ corresponds to the 10th percentile.

$x = \mu + z\sigma = 1050 + (-1.28)(150) = 1050 - 192 = 858$.

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10 [Intermediate] • Target: Assessing normality

Explain how a normal probability plot (or normal quantile plot) can be used to assess whether a data set is approximately normal, and what a roughly straight-line pattern indicates.

A normal probability plot compares the ordered data values to what would be expected if the data were exactly normal. If the plotted points fall close to a straight line, the data is approximately normal. Systematic curvature or bending away from a straight line (such as an S-shape or clear bends at the ends) suggests the data deviates from normality — for example, skewness or heavy tails.

11 [Intermediate] • Target: Applying the normal distribution to real data

Pregnancy lengths are approximately normal with mean $\mu = 266$ days and standard deviation $\sigma = 16$ days. Find the z-score for a pregnancy lasting 250 days.

$z = \frac{250 - 266}{16} = \frac{-16}{16} = -1$.

12 [Intermediate] • Target: Applying the normal distribution to real data

Using the same pregnancy-length distribution and the 68-95-99.7 rule, find the approximate percentage of pregnancies lasting between 234 and 298 days.

$234 = 266 - 32 = \mu - 2\sigma$ and $298 = 266 + 32 = \mu + 2\sigma$, so this range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of pregnancies fall in this range.

13 [Intermediate] • Target: Combining two independent normal random variables

A vending machine dispenses a soda whose volume is normally distributed with mean 355 mL and standard deviation 4 mL, independently of a snack bag whose weight is normally distributed with mean 50 g and standard deviation 2 g. If T represents the combined "total" formed by adding the soda volume (in mL) and snack weight (in g) as a single number, find the mean of T. (This is a simplified illustrative combination, not a real physical unit.)

For independent normal random variables, means add directly: $\mu_T = 355 + 50 = 405$.

14 [Intermediate] • Target: Combining two independent normal random variables

Battery lifespans are normally distributed with mean 500 hours and standard deviation 40 hours. Find the z-score for a battery lasting 460 hours, and describe its relative position.

$z = \frac{460 - 500}{40} = \frac{-40}{40} = -1$. This battery's lifespan is exactly one standard deviation below the mean.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging]** • **Target: Combining two independent normal random variables** • **Cross-Lesson: 1.02 Z-Scores & Linear Transformations**

A company manufactures resistors with resistance values approximately normal, mean 100 ohms and standard deviation 5 ohms. A resistor is considered defective if its resistance is below 90 ohms or above 110 ohms. Using the 68-95-99.7 rule, estimate the percentage of resistors that are defective, and explain your reasoning step by step.

$90 = 100 - 2(5) = \mu - 2\sigma$ and $110 = 100 + 2(5) = \mu + 2\sigma$, so the acceptable range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of resistors fall within this range (not defective), leaving approximately 5% defective (split as roughly 2.5% below 90 ohms and 2.5% above 110 ohms).

16 **[Challenging]** • **Target: Combining two independent normal random variables** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

Two students are comparing their standardized test scores from different tests. Student X scored 78 on a test with mean 70 and standard deviation 8. Student Y scored 165 on a different test with mean 150 and standard deviation 12. Using z-scores, determine who performed relatively better compared to their own test's distribution of test-takers, showing full reasoning.

Student X: $z = \frac{78 - 70}{8} = \frac{8}{8} = 1$. Student Y: $z = \frac{165 - 150}{12} = \frac{15}{12} = 1.25$. Since $1.25 > 1$, Student Y performed relatively better compared to their own test's distribution of test-takers, even though the raw score gap looks different for each test.

17 **[Challenging]** • **Target: Properties of the normal distribution** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

The distribution of cholesterol levels in a population is approximately normal with $\mu = 190$ mg/dL and $\sigma = 24$ mg/dL. (a) What proportion of individuals have cholesterol levels above 226 mg/dL? (b) What cholesterol level marks the 90th percentile?

(a) Proportion Above 226: Standardize: $z = \frac{226 - 190}{24} = \frac{36}{24} = 1.50$. Using standard normal tables: $P(Z > 1.50) = 1 - 0.9332 = \mathbf{0.0668}$ (6.68%). (b) 90th Percentile: Find z^* such that $P(Z < z^*) = 0.90$, which gives $z^* \approx 1.282$. Unstandardize: $x = \mu + z^*\sigma = 190 + 1.282(24) = 190 + 30.77 = \mathbf{220.77}$ mg/dL.

18 **[Challenging]** • **Target: The 68-95-99.7 rule** • **Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A basketball player makes 75% of free throws. In a game, she shoots 12 free throws. Assuming attempts are independent: (a) Find the probability she makes exactly 9 free throws. (b) Find the mean and standard deviation of made free throws.

(a) Binomial PMF: $X \sim B(n = 12, p = 0.75)$, $k = 9$: $P(X = 9) = \binom{12}{9}(0.75)^9(0.25)^3 = 220 \times 0.075085 \times 0.015625 \approx \mathbf{0.2581}$. (b) Mean & Standard Deviation: $\mu = np = 12(0.75) = \mathbf{9.0}$ free throws.

$\sigma = \sqrt{np(1-p)} = \sqrt{12(0.75)(0.25)} = \sqrt{2.25} = \mathbf{1.5}$ free throws.

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19 [Challenging] • Target: Standardizing and using the normal distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

[Unit 1 · Exploring One-Variable & Two-Variable Data · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 [Challenging] • Target: Visualizing the standard normal distribution • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.

1. Define the difference variable $W = X - Y$. The condition $X > Y$ is equivalent to $W > 0$. 2. Since X and Y are independent normals, W is normally distributed: Mean: $\mu_W = E[X] - E[Y] = 50 - 45 = 5$. Variance: $\sigma_W^2 = \text{Var}(X) + \text{Var}(Y) = 16 + 9 = 25 \Rightarrow \sigma_W = 5$. Thus $W \sim N(5, 25)$. 3. Compute $P(W > 0)$: $z = \frac{0 - 5}{5} = -1.0$. $P(W > 0) = P(Z > -1.0) = P(Z < 1.0) = \Phi(1.0) \approx 0.8413 \approx 0.841$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Properties of the normal distribution

A normal distribution has mean 64 and standard deviation 9. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. $(46, 82)$ ✓

b. $(37, 91)$

c. $(46, 81)$

d. $(47, 82)$

The middle 95% range spans $\mu \pm 2\sigma = 64 \pm 2(9) = 64 \pm 18$, so from 46 to 82: $(46, 82)$.

22 [Foundational] • Target: The 68-95-99.7 rule

A normal distribution has mean $\mu = 101$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 103$, to two decimal places.

a. 69.15% ✓

b. 72.15%

c. 66.15%

d. 30.85%

$z = \frac{103 - 101}{4} = 0.50$. Using the standard normal distribution, $P(Z < 0.50) \approx 0.6915$, or about 69.15%.

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23 [Foundational] • Target: Standardizing and using the normal distribution

A normal distribution has mean $\mu = 144$ and standard deviation $\sigma = 16$. Find the approximate percentage of values between $x_1 = 136$ and $x_2 = 152$, to two decimal places.

a. 69.15%

b. 34.29%

c. 42.29%

d. 38.29% ✓

$z_1 = \frac{136 - 144}{16} = -0.50$. $z_2 = \frac{152 - 144}{16} = 0.50$. $P(-0.50 < Z < 0.50) \approx 0.3829$, or about 38.29%.

24 [Foundational] • Target: Visualizing the standard normal distribution

Scores on an exam are approximately normal with mean $\mu = 390$ and standard deviation $\sigma = 15$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.

a. 365

b. 405

c. 425

d. 415 ✓

$x = \mu + z\sigma = 390 + (1.645)(15) = 415$.

25 [Foundational] • Target: Working backward: finding values from percentiles

A company manufactures parts with a measurement approximately normal, mean 152 and standard deviation 5. A part is considered defective if its measurement is below 142 or above 162 (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 0.3%

b. 5% ✓

c. 32%

d. 2.5%

$142 = \mu - 2\sigma$ and $162 = \mu + 2\sigma$, so the acceptable range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of parts fall within this range, leaving approximately 5% defective (roughly 2.5% in each tail).

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26 [Foundational] • Target: Assessing normality

A data set has $Q_1 = 17$ and $Q_3 = 35$. Using the 1.5×IQR rule, which of the following values would be classified as an outlier?

a. 35

b. **-14 ✓**

c. 17

d. -9

IQR $= 35 - 17 = 18$. Lower fence $= 17 - 1.5(18) = -10$. Upper fence $= 35 + 1.5(18) = 62$. The value -14 is below the lower fence, so it is an outlier.

27 [Foundational] • Target: Applying the normal distribution to real data

A researcher is designing an experiment to test the effectiveness of a new fertilizer on plant growth. They have a total of 50 experimental plots available. The experiment will involve two treatments: the new fertilizer (Treatment A) and a standard fertilizer (Treatment B). To ensure fair comparison and control for confounding variables, they decide to use a randomized block design. The plots are divided into 5 blocks based on soil quality, with each block containing 10 plots. Within each block, the plots will be randomly assigned to either Treatment A or Treatment B. If the number of plots assigned to Treatment A within the first block is x, and the total number of plots assigned to Treatment A across all 5 blocks is 25, and the proportion of plots assigned to Treatment A in block 1 is $\frac{1}{5}$ of the total plots in block 1, what is the value of x?

a. **3 ✓**

b. 4

c. 5

d. 6

The problem describes a randomized block design. There are 5 blocks, and each block has 10 plots, for a total of 50 plots. The experiment uses two treatments: Treatment A (new fertilizer) and Treatment B (standard fertilizer). The total number of plots assigned to Treatment A across all 5 blocks is 25. The question states that the proportion of plots assigned to Treatment A in block 1 is $\frac{1}{5}$ of the total plots in block 1. The number of plots in block 1 is 10. Therefore, the number of plots assigned to Tre.

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28 [Foundational] • Target: Combining two independent normal random variables

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

a. 0.980 ✓

b. 0.900

c. 0.810

d. 0.990

1. **Margin of error for each confidence interval:** For a symmetric 90% confidence interval with known σ, the critical value is $z = 1.645$. Each interval has half-width: $E = 1.645\frac{\sigma}{\sqrt{n}}$ The two intervals are $I_1 = [\bar{X}_1 - E, \bar{X}_1 + E]$ and $I_2 = [\bar{X}_2 - E, \bar{X}_2 + E]$. 2. **Condition for interval overlap:** Two intervals of identical half-width E overlap if and only if the distance between their centres does not exceed the sum of their half-widths.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Properties of the normal distribution

A researcher is designing a survey to understand student opinions on campus dining. The researcher decides to select students from the student directory using a systematic random sampling method. If the student directory has a total of 1500 students, and the researcher decides to select every 10th student, starting with a randomly selected student between 1 and 10, how many students will be in the sample?

a. 150 ✓

b. 10

c. 15

d. 1500

The sample size is determined by dividing the total population size by the sampling interval. The total population size is 1500 students, and the sampling interval is 10. Therefore, the number of students in the sample is $\frac{1500}{10} = 150$. The starting point between 1 and 10 ensures that the selection is random, but it does not affect the total number of students sampled.

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30 [Intermediate] • Target: The 68-95-99.7 rule

A data set contains the values 4, 13, 14, 23, 30, 33, 35. What is the mean of this data set?

a. 23

b. **21.71 ✓**

c. 23.89

d. 19.71

Mean $= \frac{\sum x_i}{n} = \frac{152}{7} = 21.71.$

31 [Intermediate] • Target: Standardizing and using the normal distribution

A data set contains the values 22, 25, 41, 51, 54, 64. What is the mean of this data set (rounded to 2 decimal places)?

a. 48.83

b. 38.83

c. **42.83 ✓**

d. 45.83

Mean $= \frac{\sum x_i}{n} = \frac{257}{6} = 42.83.$

32 [Intermediate] • Target: Visualizing the standard normal distribution

A researcher wants to investigate the relationship between the type of pet owned (Dog, Cat, Fish) and the preferred music genre (Rock, Pop, Classical). They collect data and present it in a two-way table. If the table has 3 rows and 3 columns, what is the total number of cells in the table that represent a combination of pet ownership and music genre?

a. 6

b. 7

c. **9 ✓**

d. 16

A two-way table is used to display the frequency distribution of two categorical variables. The number of rows corresponds to the categories of one variable, and the number of columns corresponds to the categories of the other variable. The total number of cells in the table is the product of the number of rows and the number of columns. In this case, the number of rows is 3 (representing Dog, Cat, Fish) and the number of columns is 3 (representing Rock, Pop, Classical). Total number of cells = Number of rows.

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33 [Intermediate] • Target: Working backward: finding values from percentiles

A researcher is investigating the relationship between study habits and exam performance. They categorize students into two groups: 'Regular Study' and 'Cramming'. They also categorize exam performance into 'Pass' and 'Fail'. The data collected is presented in a two-way table. If the table has 2 rows representing study habits and 3 columns representing exam performance categories, what is the total number of cells in this two-way table?

a. 4

b. 5

c. **6** ✓

d. 3

A two-way table displays the frequencies of two categorical variables. The number of rows represents the number of categories for the first variable, and the number of columns represents the number of categories for the second variable. To find the total number of cells in the table, we multiply the number of rows by the number of columns. Number of rows = 2 Number of columns = 3 Total number of cells = Number of rows $\times$ Number of columns = $2 \times 3 = 6$.

34 [Intermediate] • Target: Assessing normality

The IQR (interquartile range) equals:

a. $Q_3 - Q_1$ ✓

b. $Q_1 - Q_3$

c. Max – Min

d. $Q_2 - Q_1$

Applying Unit 1 · Exploring One-Variable & Two-Variable Data principles, IQR=Q3–Q1. Therefore, the verified correct option is $Q_3 - Q_1$.

35 [Intermediate] • Target: Applying the normal distribution to real data

From a randomly ordered list of 1854 customers, a researcher selects every 21th person after choosing a random starting point. What sampling method is this?

a. Convenience sample

b. **Systematic sample** ✓

c. Stratified random sample

d. Simple random sample (SRS)

Selecting individuals at fixed intervals (every 21th) after a random start defines a systematic sample.

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36 [Intermediate] • Target: Combining two independent normal random variables

From a randomly ordered list of 1002 employees, a researcher selects every 14th person after choosing a random starting point. What sampling method is this?

a. Cluster sample

b. **Systematic sample ✓**

c. Voluntary response sample

d. Stratified random sample

Selecting individuals at fixed intervals (every 14th) after a random start defines a systematic sample.

37 [Intermediate] • Target: Properties of the normal distribution

From a randomly ordered list of 885 voters, a researcher selects every 24th person after choosing a random starting point. What sampling method is this?

a. Simple random sample (SRS)

b. Voluntary response sample

c. Convenience sample

d. **Systematic sample ✓**

Selecting individuals at fixed intervals (every 24th) after a random start defines a systematic sample.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The 68-95-99.7 rule • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

From a randomly ordered list of 610 students, a researcher selects every 18th person after choosing a random starting point. What sampling method is this?

a. Simple random sample (SRS)

b. **Systematic sample ✓**

c. Convenience sample

d. Stratified random sample

Selecting individuals at fixed intervals (every 18th) after a random start defines a systematic sample.

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39 [Challenging] • Target: Standardizing and using the normal distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

From a randomly ordered list of 1390 athletes, a researcher selects every 7th person after choosing a random starting point. What sampling method is this?

a. Stratified random sample

b. **Systematic sample ✓**

c. Simple random sample (SRS)

d. Voluntary response sample

Selecting individuals at fixed intervals (every 7th) after a random start defines a systematic sample.

40 [Challenging] • Target: Visualizing the standard normal distribution • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

From a randomly ordered list of 1515 patients, a researcher selects every 6th person after choosing a random starting point. What sampling method is this?

a. Convenience sample

b. **Systematic sample ✓**

c. Simple random sample (SRS)

d. Stratified random sample

Selecting individuals at fixed intervals (every 6th) after a random start defines a systematic sample.

41 [Challenging] • Target: Working backward: finding values from percentiles • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A data set (already sorted, n=9) is 1, 7, 10, 11, 25, 34, 43, 46, 51. What is the five-number summary?

a. Min = 1, Q1 = 10, Median = 51, Q3 = 43, Max = 25

b. Min = 1, Q1 = 43, Median = 25, Q3 = 10, Max = 51

c. **Min = 1, Q1 = 10, Median = 25, Q3 = 43, Max = 51 ✓**

d. Min = 51, Q1 = 10, Median = 25, Q3 = 43, Max = 1

With n=9: Min=1, Q1 (position 3) =10, Median (position 5) =25, Q3 (position 7) =43, Max=51.

42 [Challenging] • Target: Assessing normality • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

An outlier in a boxplot is typically beyond:

a. **1.5 × IQR from Q1 or Q3 ✓**

b. 2 × IQR

c. 3 standard deviations

d. The median

Standard rule: more than 1.5×IQR from quartiles.

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43 [Challenging] • Target: Applying the normal distribution to real data • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A regression line predicts $\hat{y} = 41.85$ for $x = 25$, while the observed value is $y = 39.1$. What is the residual for this point?

a. 3.25

b. **-2.75 ✓**

c. -0.75

d. -6.75

Residual $= y - \hat{y} = 39.1 - 41.85 = -2.75$.

44 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.01 Measures of Center & Spread

Which test compares two independent population means?

a. One-sample z-test

b. Paired t-test

c. **Two-sample t-test ✓**

d. Chi-square test

Two independent groups → two-sample t-test.

45 [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A randomised block design blocks subjects by age group into 3 blocks of 44 subjects each, with random assignment to treatments within each block. What is the total sample size?

a. 117

b. 107

c. 142

d. **132 ✓**

Total $= 3 \times 44 = 132$ subjects.

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